

Problem 4.2

At what annual rate of simple interest will \$500 accumulate to \$615 in $2\frac{1}{2}$ years?

Solution

For simple interest,

$$A = P(1 + it).$$

Substituting the given values,

$$615 = 500 \left(1 + i \left(\frac{5}{2} \right) \right).$$

Dividing by 500,

$$\frac{615}{500} = 1 + i \left(\frac{5}{2} \right).$$

Therefore,

$$i = \left(\frac{615}{500} - 1 \right) \frac{2}{5}.$$

Hence,

$$i = 0.092.$$

So,

$$i = 9.2\%.$$